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Task 1: Data Analysis Using Pandas and Matplotlib

1. Introduction

Data analysis is the process of examining data to obtain useful information. In Python, libraries such as Pandas and Matplotlib are commonly used for handling data and creating graphical representations. In this task, a CSV file containing student marks is analyzed to understand subject-wise performance using basic statistical methods and visualizations.

2. Objective

The objectives of this task are:

- To read data from a CSV file using Pandas
 - To calculate the average marks of selected subjects
 - To visualize the data using different graphs
 - To analyze and interpret the results obtained from the data
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3. Tools and Technologies Used

- **Language:** Python
 - **Libraries:**
 - Pandas – for data reading and manipulation
 - Matplotlib – for graphical representation
 - Seaborn – for correlation visualization
 - **IDE:** Visual Studio Code
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4. Dataset Description

The dataset used in this task is named student.csv. It contains marks of students in three subjects:

- Maths
- Science
- English

Each record represents the marks obtained by a student in the respective subjects.

5. Methodology

1. The CSV file was loaded using the `read_csv()` function of Pandas.
 2. The dataset was displayed to understand its structure.
 3. The average marks of each subject were calculated using the `mean()` function.
 4. A bar chart was used to show subject-wise average marks.
 5. A scatter plot was used to observe the relationship between Maths and Science marks.
 6. A heatmap was generated to identify the correlation between the subjects.
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6. Results and Observations

- The average marks are highest in **English**, followed by **Science**, and lowest in **Maths**.
 - The scatter plot shows a positive relationship between Maths and Science marks.
 - The heatmap indicates a positive correlation among all subjects, which shows that student performance is consistent across subjects.
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7. Conclusion

This task helped in understanding basic data analysis using Python. Pandas was used for data handling, and Matplotlib was used for visualization. The graphs provided a clear understanding of student performance in different subjects. This task improved practical knowledge of data analysis and visualization techniques.

8. Output

- Python program (`data_analysis.py`)
- CSV dataset (`student.csv`)
- Output graphs (bar chart, scatter plot, heatmap)